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Getting Started with Microsoft Power BI





Course Agenda



Module 1: Power BI Visualization Essentials



Module 2: Power BI Service & Collaboration



About us:



Nicolás Lagreste Zucchini 

✉ nico@analyticmood.com

🌐 www.analyticmood.com

analytic mood

- **Co-founder of Analytic Mood** — Leading data solutions and business strategies.
- **Experienced Data Expert and Business Analyst** — Extensive experience with NTT Data, Kabel, BI Applications, Grupo Solutio and Analytic Mood.
- **Microsoft Power BI Lead** — Oversees the full development cycle from design to deployment.
- **Power BI Community Speaker** — Presented at the latest Power BI Summit, engaging with industry peers.



María Florencia Hourcouripe 

✉ flor@analyticmood.com

🌐 www.analyticmood.com

analytic mood

- **Co-founder of Analytic Mood** — Pioneering data-driven solutions and strategic innovations.
- **Educational Background** — Holds a Master's degree in "Business Intelligence & Technological Innovation" from EAE Business School.
- **Consultant & Trainer in Microsoft Power BI** — Expert in guiding professionals through Power BI applications and analytics.
- **Power BI Community Contributor** — Recently spoke at the Power BI Summit, sharing expertise with fellow data enthusiasts.

Initial Setup

Download the shared **.ZIP** file from:

 **URL:** <https://www.analyticmood.com/en/resources>

 **Username and password will be provided in the chat.**

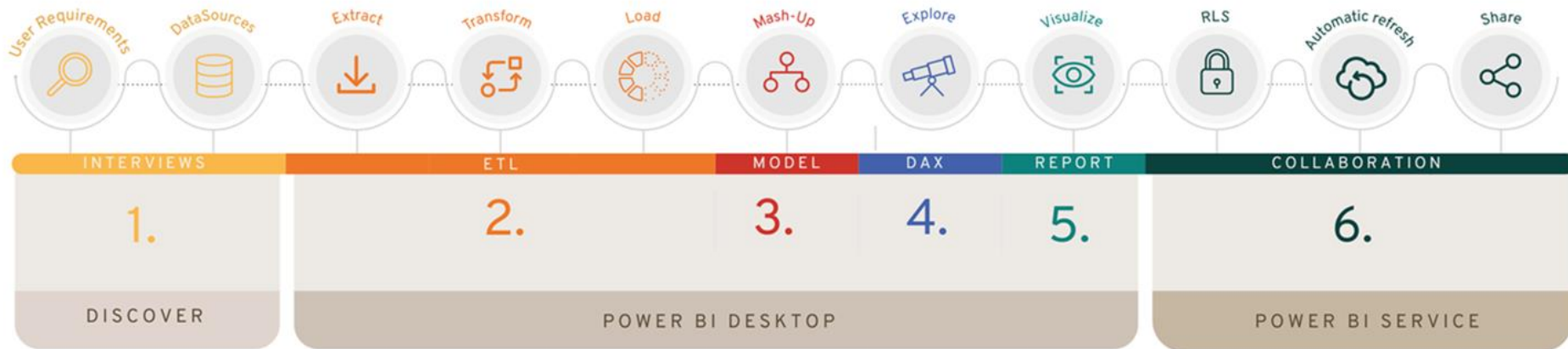


*Students should have the latest version of
Power BI Desktop installed.



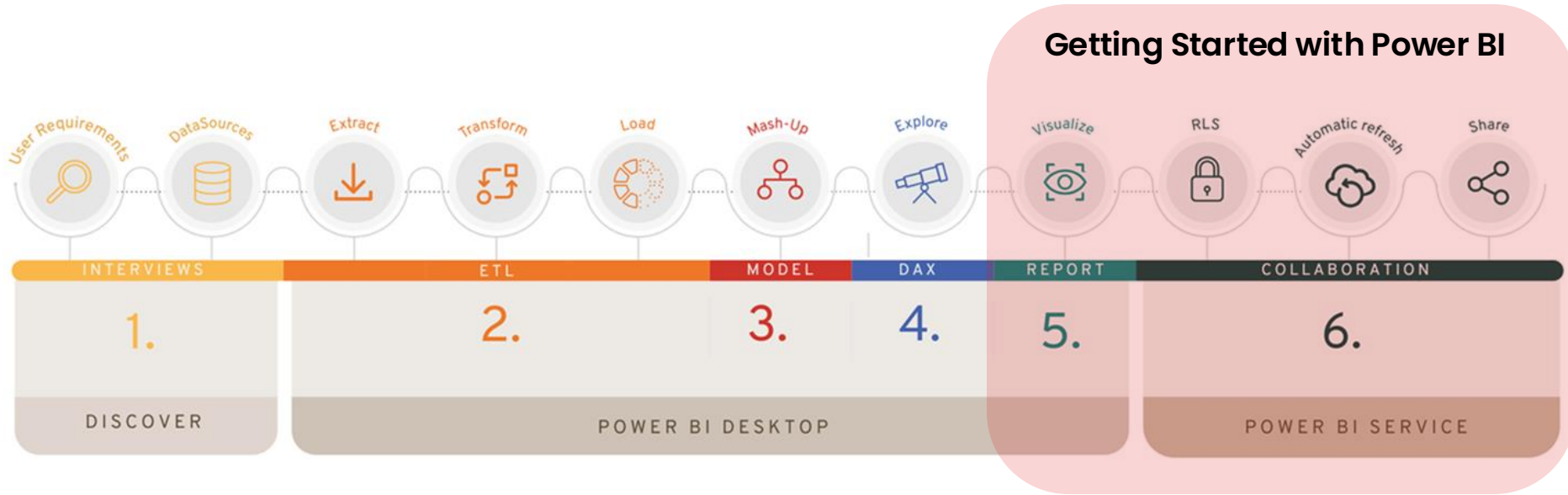


Our Method: Flow of Report Development





Our Method: Flow of Report Development



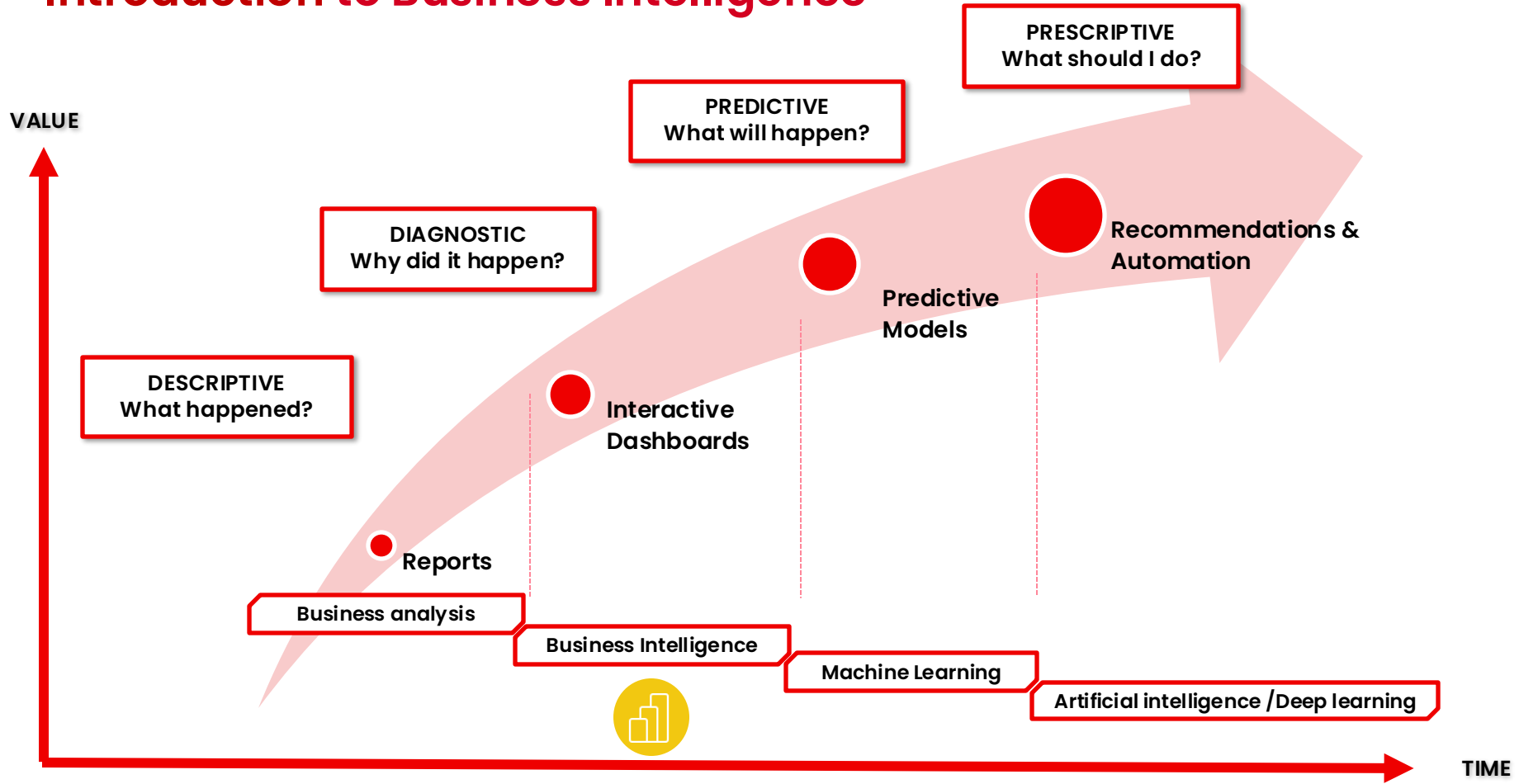


Introduction to Power BI

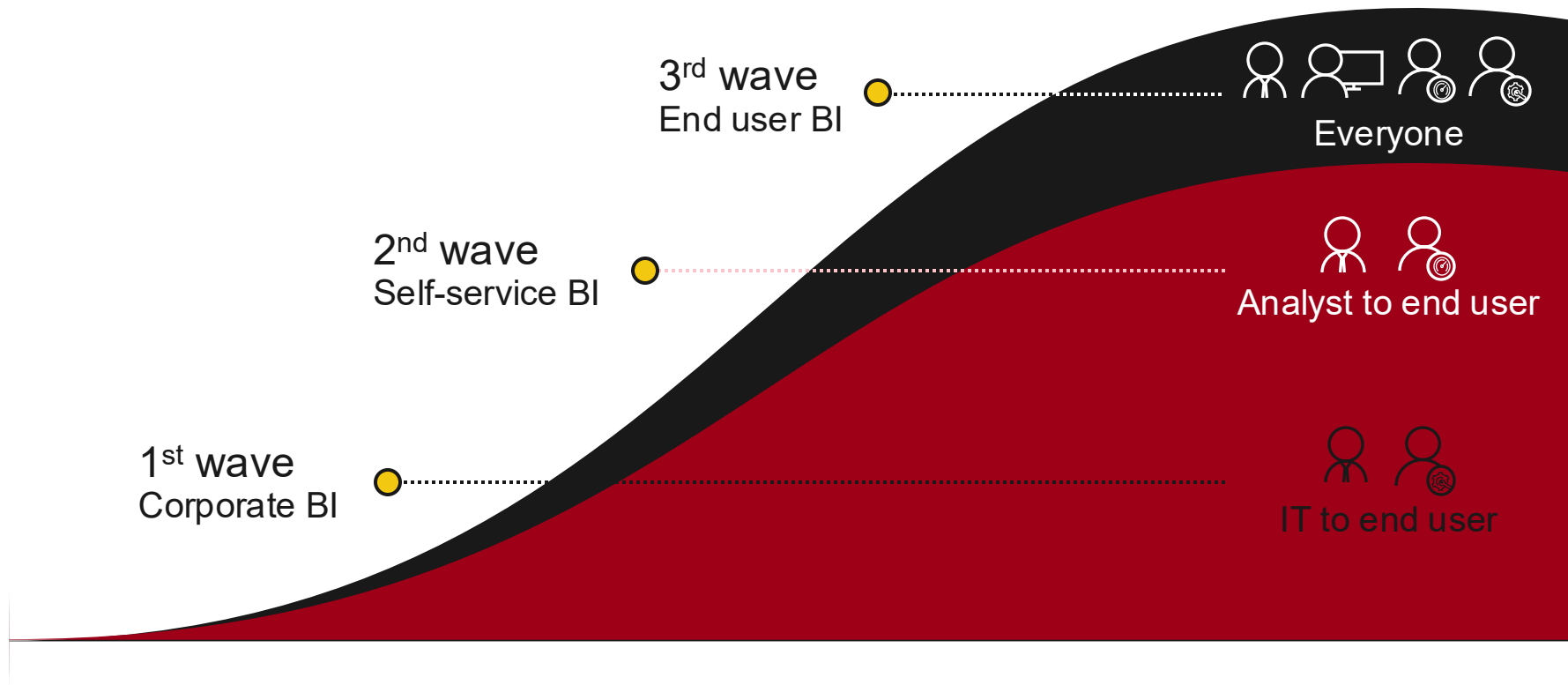




Introduction to Business Intelligence



Evolution of BI



Power BI



Create a data-driven culture with business intelligence for all

Enable everyone at every level of your organisation to make confident decisions using up-to-the-minute analytics.



Centralise your data in a trusted and secure hub

Easily create datasets from any data source and add them to the Power BI data hub to create an accessible, single source of truth for all your data.



Uncover actionable insights with visuals

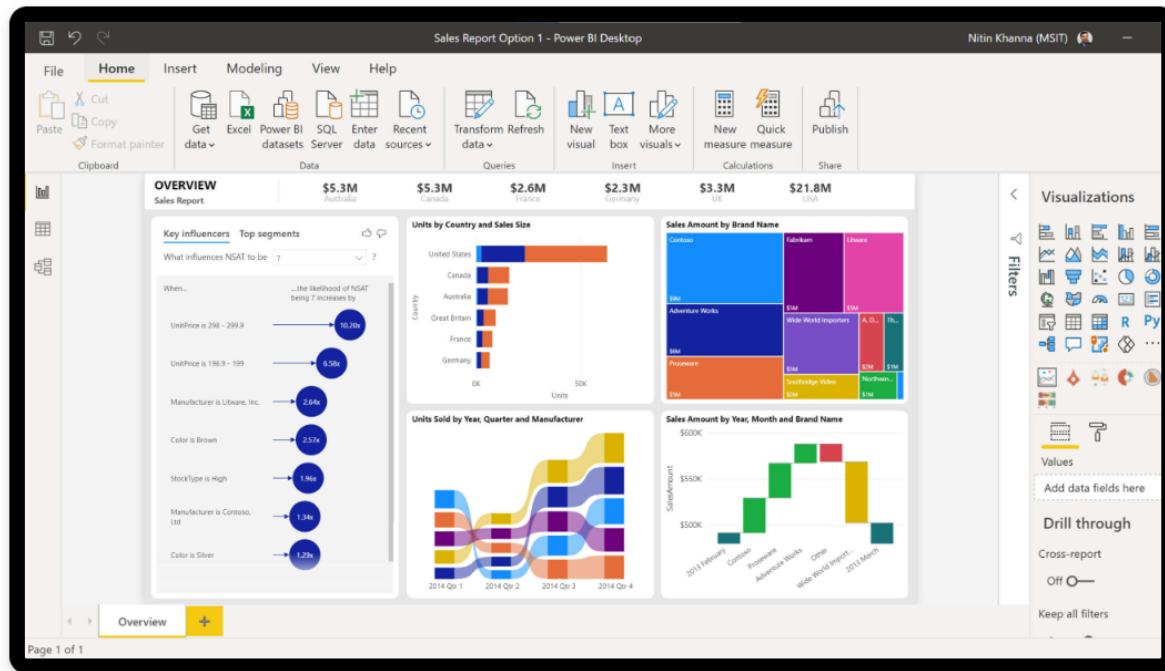
Turn your raw data into engaging visuals with industry-leading data analysis tools—powered by AI—and an approachable drag-and-drop report canvas.



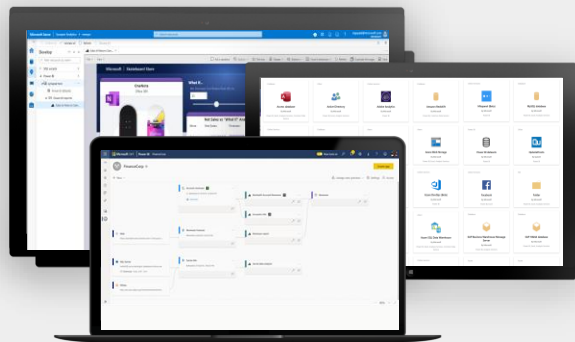
Translate insights into impact across teams

Turn insights into decisions faster by infusing insights into the apps you already use, like Microsoft 365, all while

Take a guided tour >



Microsoft Power BI



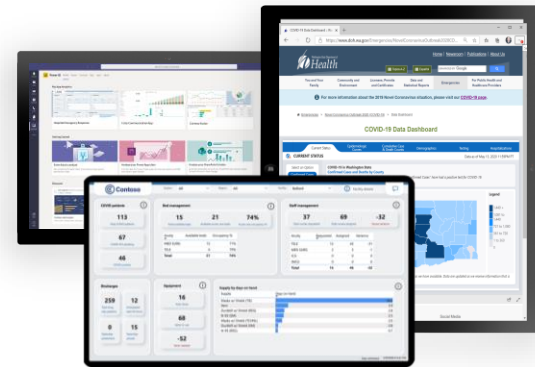
All Data

Break down data silos and
enrich your data with
intelligence



For Everyone

Create rich-interactive
data experiences with
AI infused insights



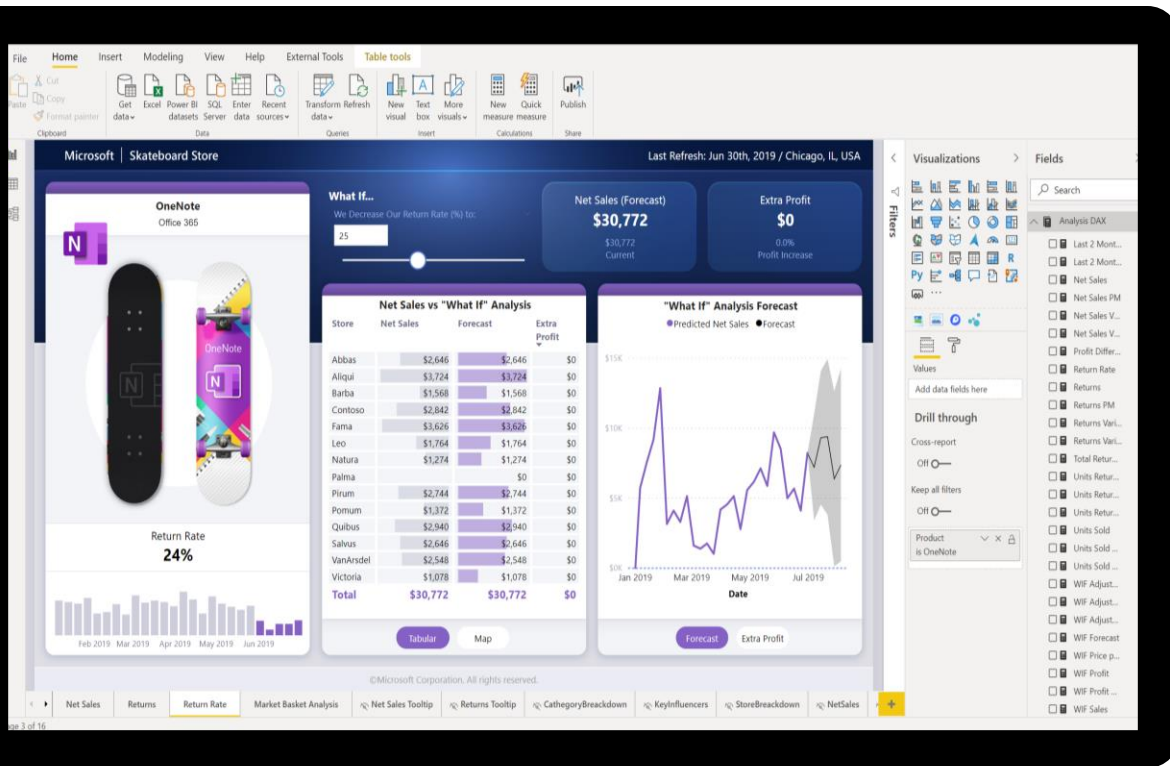
For Every Decision

Integrate trusted and secure BI
into the fabric of your
organization and apps

Power BI drives a data culture for everyone and every decision



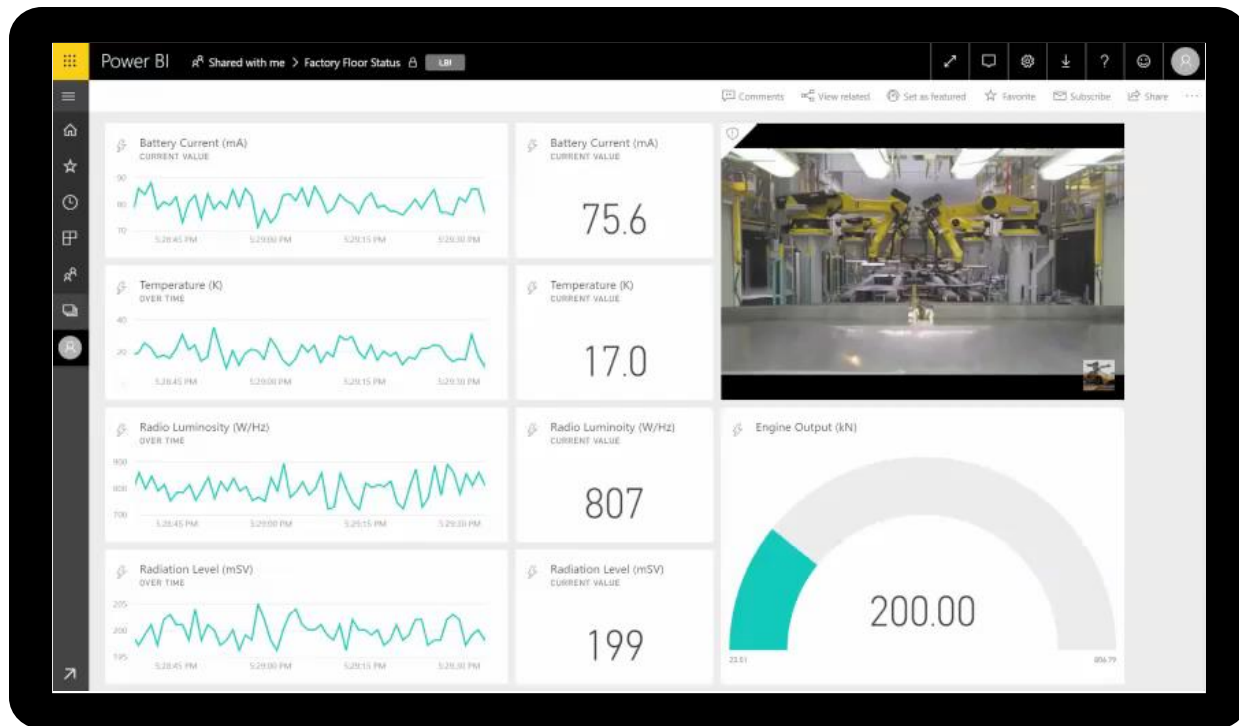
Why using Power BI?



Power BI
Desktop is free



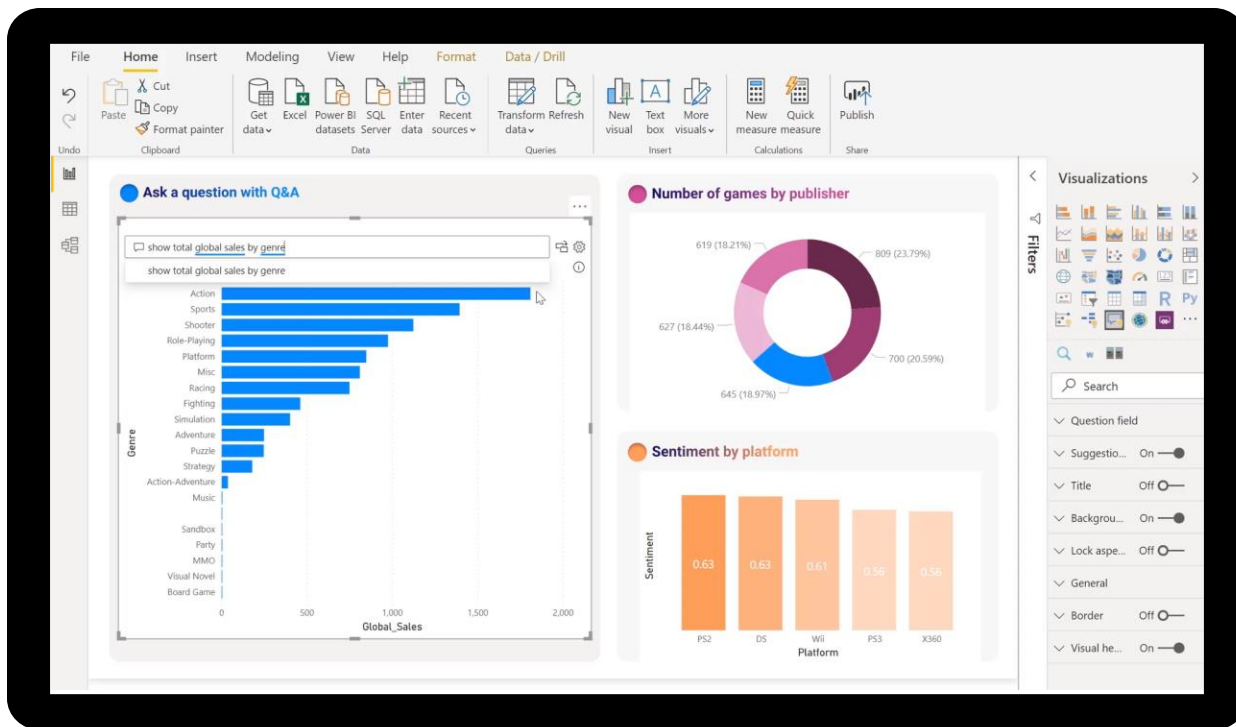
Why using Power BI?



Real time
dashboards
and interactive
reports



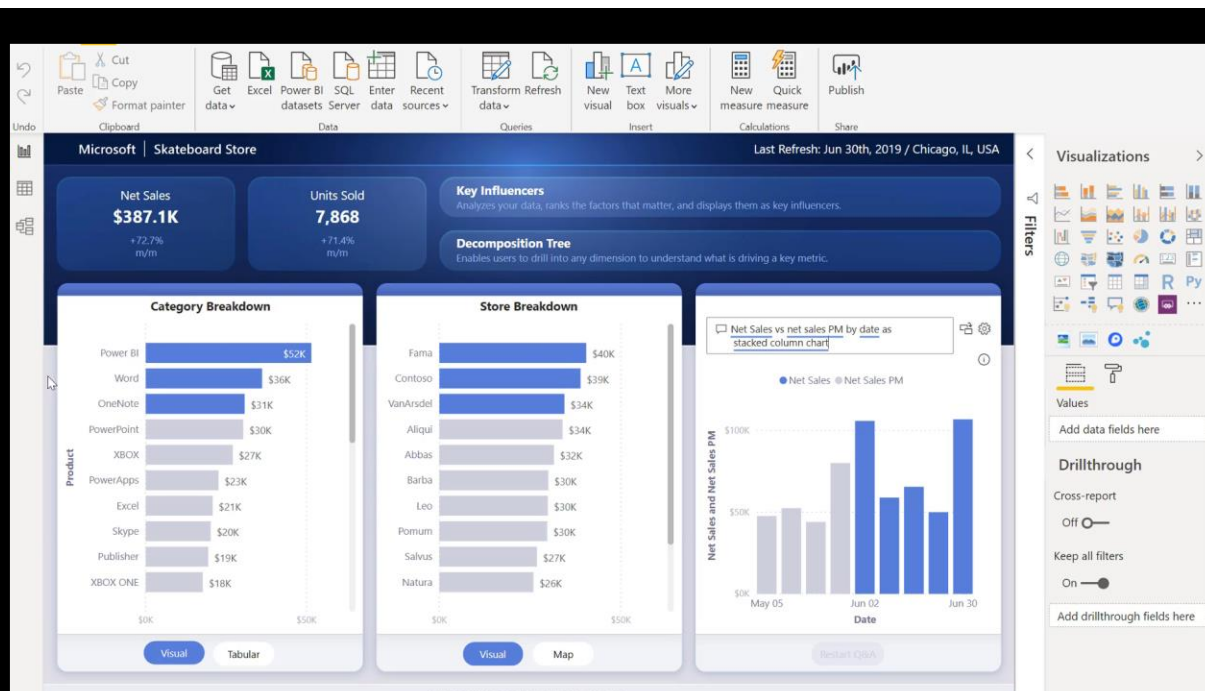
Why using Power BI?



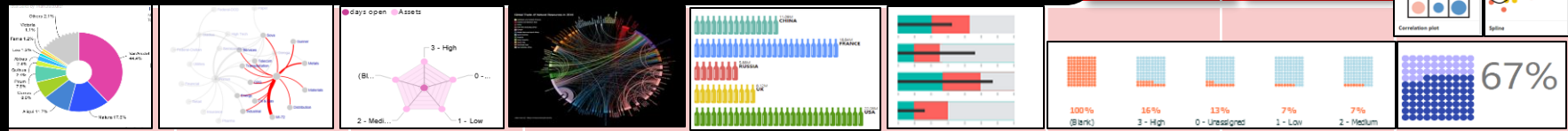
Natural
Language
query & AI
insights



Why using Power BI?

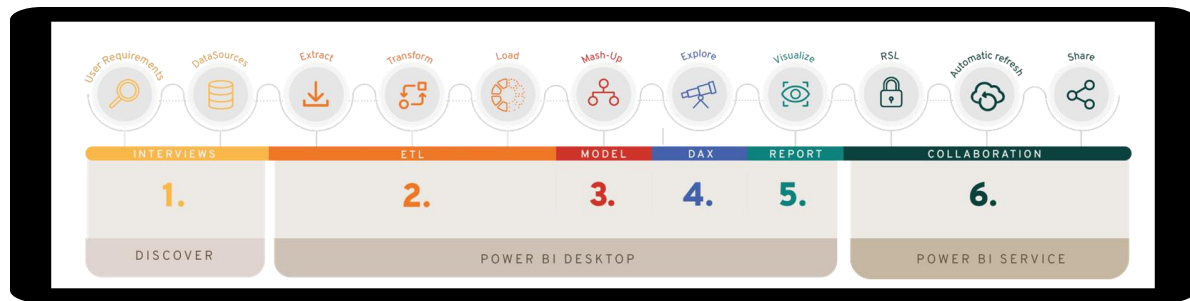


250 visuals
+ Custom visuals

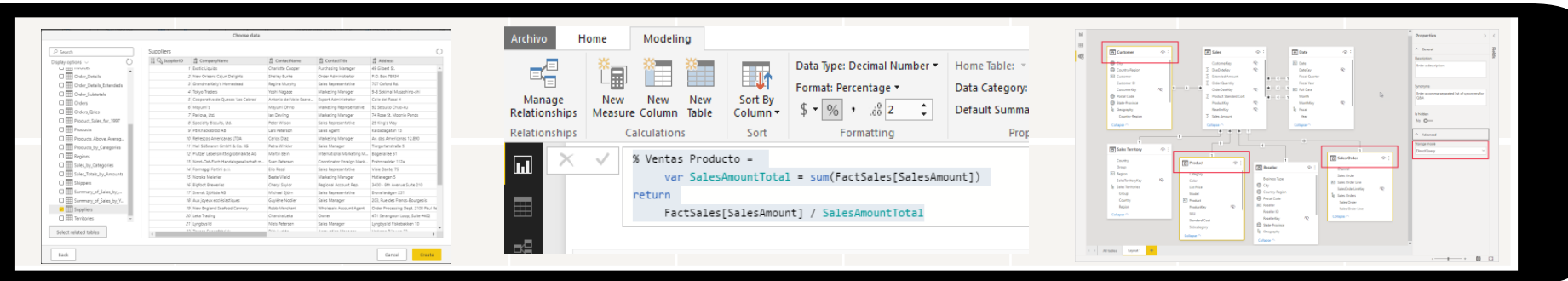




Why using Power BI?



All Business Intelligence process in one



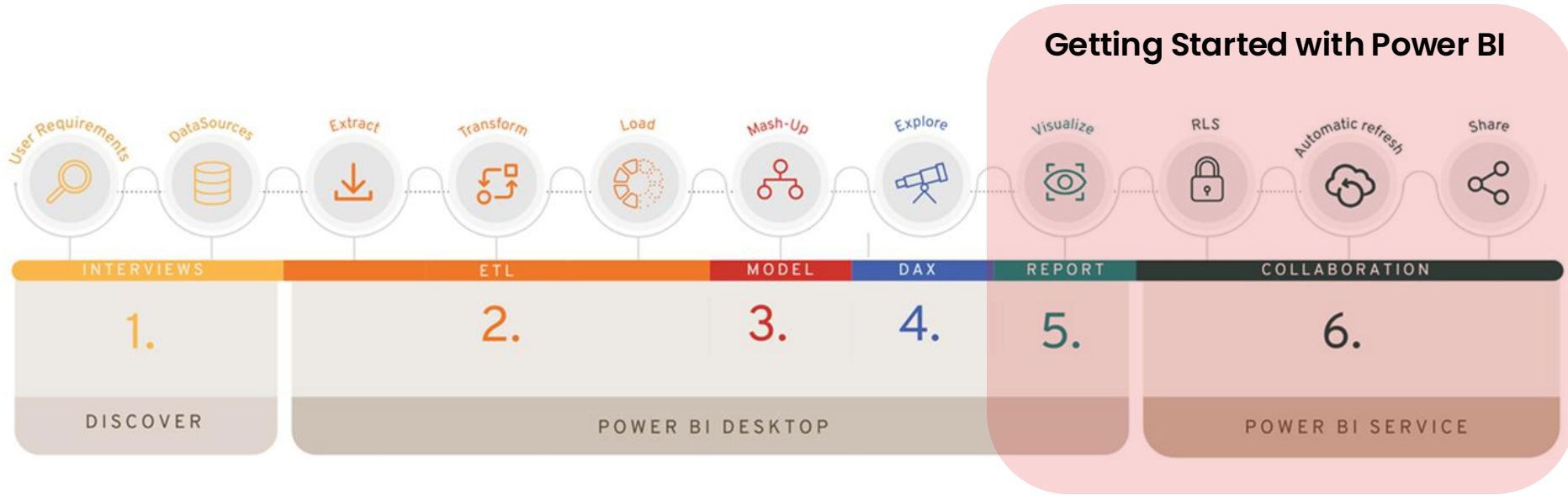
Why using Power BI?



Gartner Magic Quadrant
for “Analytics and
Business Intelligence
Platforms”



Our Method: Flow of Report Development



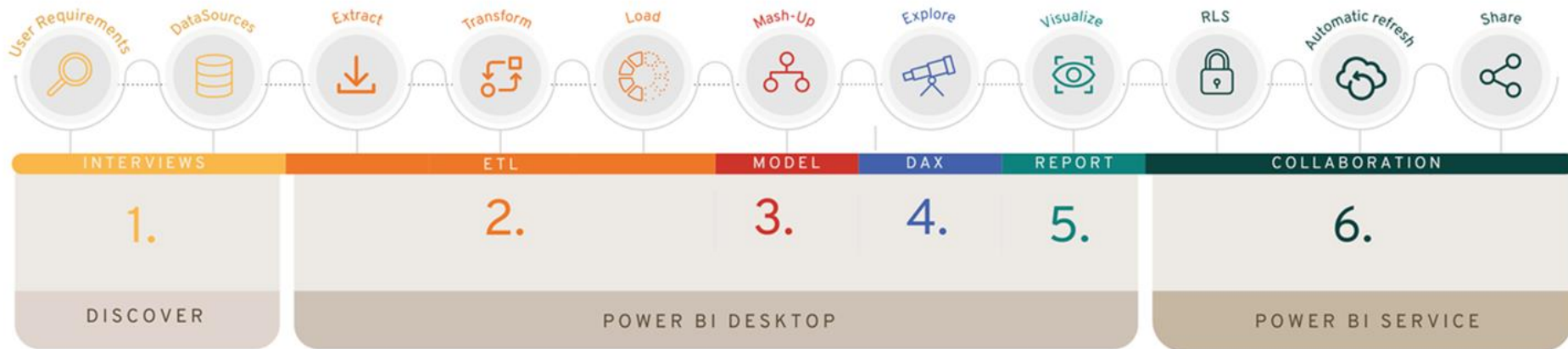


Creating a basic new report



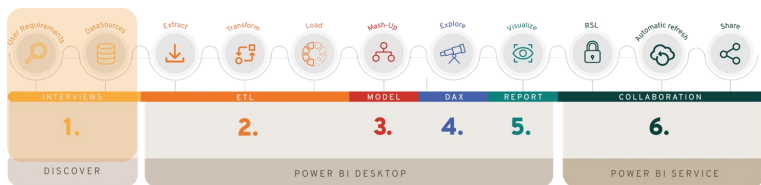


Our Method: Flow of Report Development





Interviews



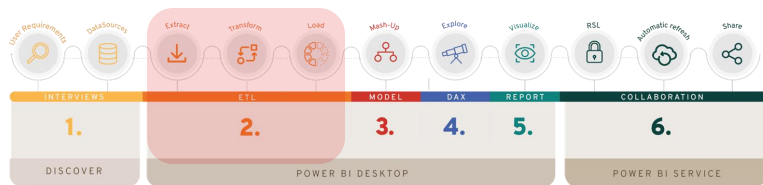
Understand processes, needs and data sources



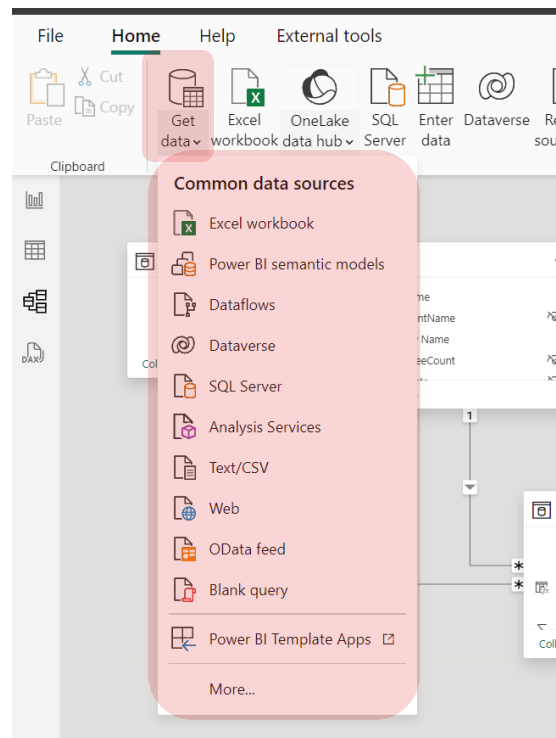
- Excel Workbook
- Text/CSV
- XML
- JSON
- Folder
- PDF
- Parquet
- SharePoint folder
- SQL Server database
- Access database
- SQL Server Analysis Services database
- Oracle database
- IBM Db2 database
- IBM Informix database (Beta)
- IBM Netezza
- MySQL database



ETL: Power Query

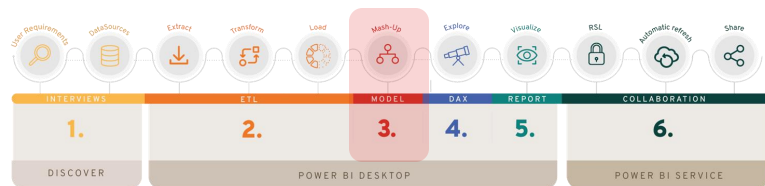


Power BI can connect to different data sources. In this PBIX we have connected to Excel, PDF and text files.





Star Model

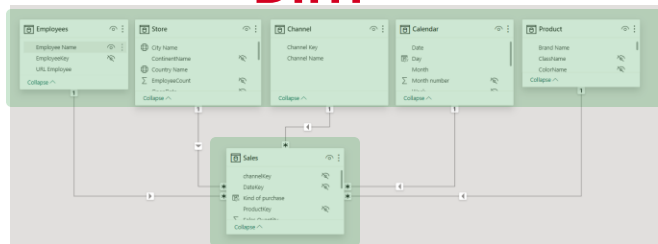


A star schema is a simple database design used to organize data efficiently.

Dim

PRODUCT_ID	PRODUCT NAME	MANUFACTURER	RANGE	TYPE
1	Apple Watch Series 7	Apple Inc.	Very Light	Smartwatch
2	Lenovo IdeaCentre AIO 520	Lenovo	Light	All-in-One PC
3	LG UltraGear 27GL850	LG	Very Light	Gaming Monitor
4	ASUS RT-AX82U	ASUS	Medium	Wi-Fi Router
5	Dell XPS 13	Dell	Light	Laptop
6	TP-Link Kasa Smart Plug	TP-Link	Light	Smart Plug
7	Sony PlayStation 5	Sony	Light	Gaming Console
8	Epson Perfection V600	Epson	Very Light	Flatbed Scanner
9	iPad Pro 12.9-inch	Apple Inc.	Medium	Tablet
10	Philips Hue White & Color	Signify	Light	Smart Bulb
11	Samsung Galaxy S22	Samsung	Very Light	Smartphone

Dim



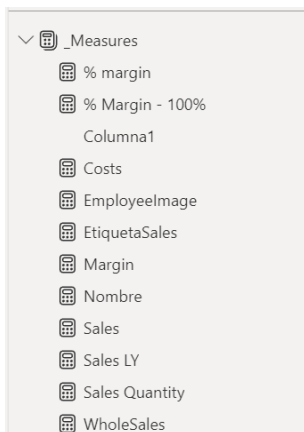
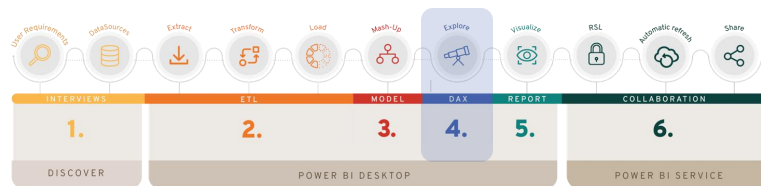
Fact

Fact

SALES AMOUNT	QUANTITY	PRODUCT_ID	EMPLOYEE_ID	DATE_ID
150	2	22	12225	20230501
210	3	17	12588	20230505
250	1	12	14558	20230509
350	5	16	13225	20230512
70	1	16	16442	20230515



Dax Measures

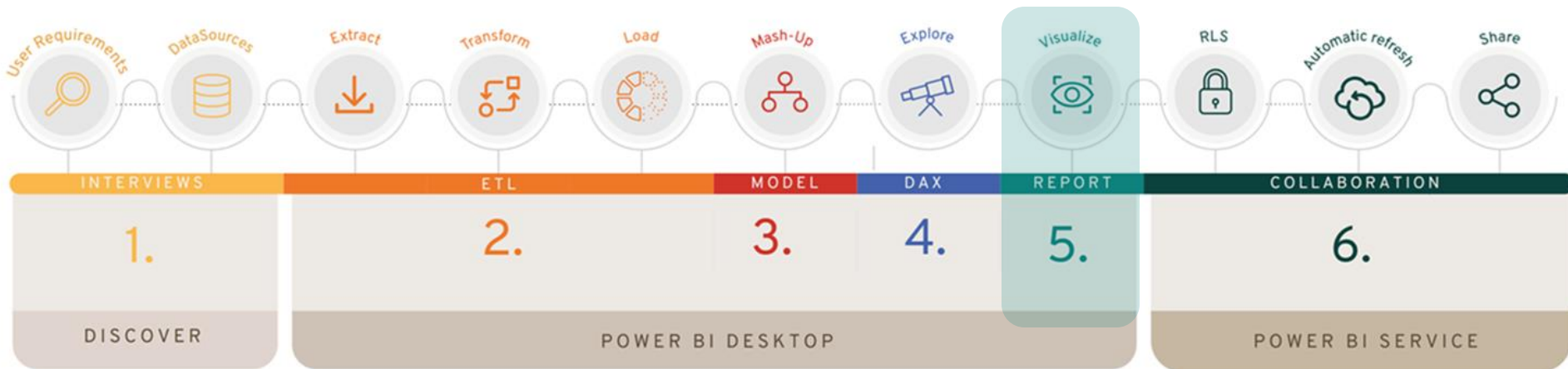


Measures in Power BI are calculations used for data analysis.

They are created using DAX formulas to perform aggregations such as sums, averages, or counts on data in your model.

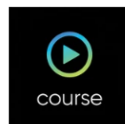


Our Method: Flow of Report Development



In this course, we'll connect to an existing model.

To learn the steps for creating a semantic model (steps 1 to 4) please visit the video course: **"Power BI Fundamentals: Building a Strong Foundation in Data Analysis"**



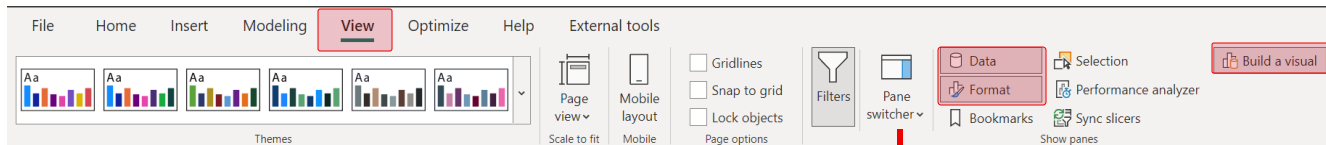
Power BI Fundamentals: Building a Strong Foundation in Data Analysis
By [Maria Florencia Hourcourpé](#) and [Nicolás Lagreste Zucchini](#)
[O'Reilly Media, Inc.](#) • February 2024

Write the [first review](#) • 5h 14m • Includes quizzes





Your first visual in Power BI

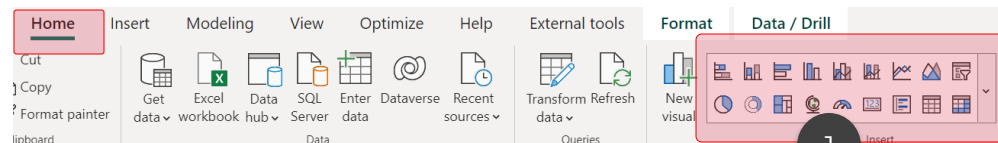


First, turn these options on.

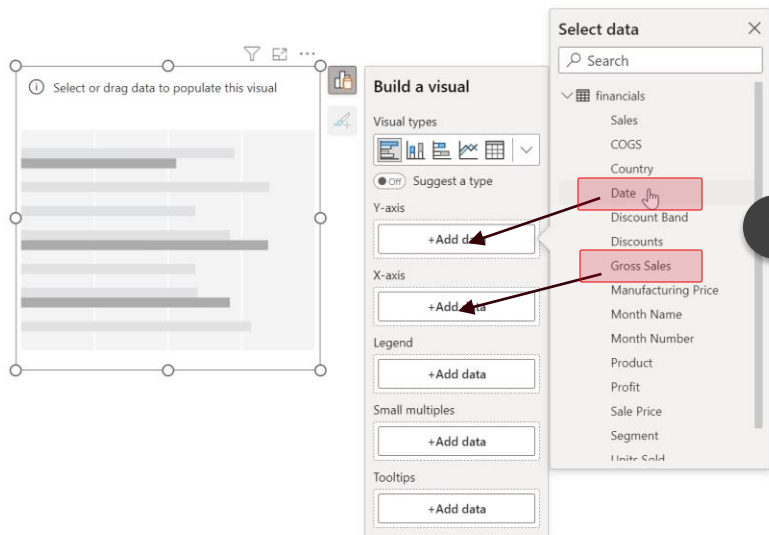
- ✓ Always show pane switcher
- ✓ Always open in a new pane
- ✓ Move Build a visual to pane switcher



Your first visual in Power BI



1
Select the visual
you want to
create



2
Then select the
fields you want
to visualize.



DEMO 1: Table, matrix, card & slicer

Settings



Home

Open

Save

Save as

Share

Get data

Import

Export

Publish

Sign in

Options and settings

About

Options and settings

Options

Data source settings

Options

GLOBAL

- Data Load
- Power Query Editor
- DirectQuery
- R scripting
- Python scripting
- Security
- Privacy
- Regional Settings
- Updates
- Usage Data
- Diagnostics
- Preview features**
- Save and Recover
- Report settings

CURRENT FILE

- Data Load
- Regional Settings
- Privacy
- Auto recovery
- Published dataset settings
- Query reduction
- Report settings

Preview features

The following features are available for you to try in this release. Preview features might change or be removed in future releases.

- ☒ Shape map visual [Learn more](#)
- ☒ Spanish language support for Q&A [Learn more](#)
- ☒ Q&A for live connected Analysis Services databases [Learn more](#)
- ☒ Connect to external datasets shared with me [Learn more](#) | [Share feedback](#)
- ☒ Modern visual tooltips [Learn more](#) | [Share feedback](#)
- ☒ Sparklines [Learn more](#)
- ☒ Metrics visual [Learn more](#)
- ☒ Quick measure suggestions [Learn more](#) | [Share feedback](#)
- ☒ Field parameters [Learn more](#)
- ☒ Horizontal fusion [Learn more](#)
- ☒ Enhanced row-level security editor [Learn more](#)
- ☒ On-object interaction [Learn more](#) | [Share feedback](#)
- ☒ Enable setting sensitivity label on exported PDF [Learn more](#)
- ☒ Dynamic format string for measures [Learn more](#)
- ☒ Save to OneDrive and SharePoint [Learn more](#)
- ☒ Share to OneDrive and SharePoint [Learn more](#)
- ☒ Power BI Project (.pbip) save option [Learn more](#)
- ☒ New card visual [Learn more](#)

OK

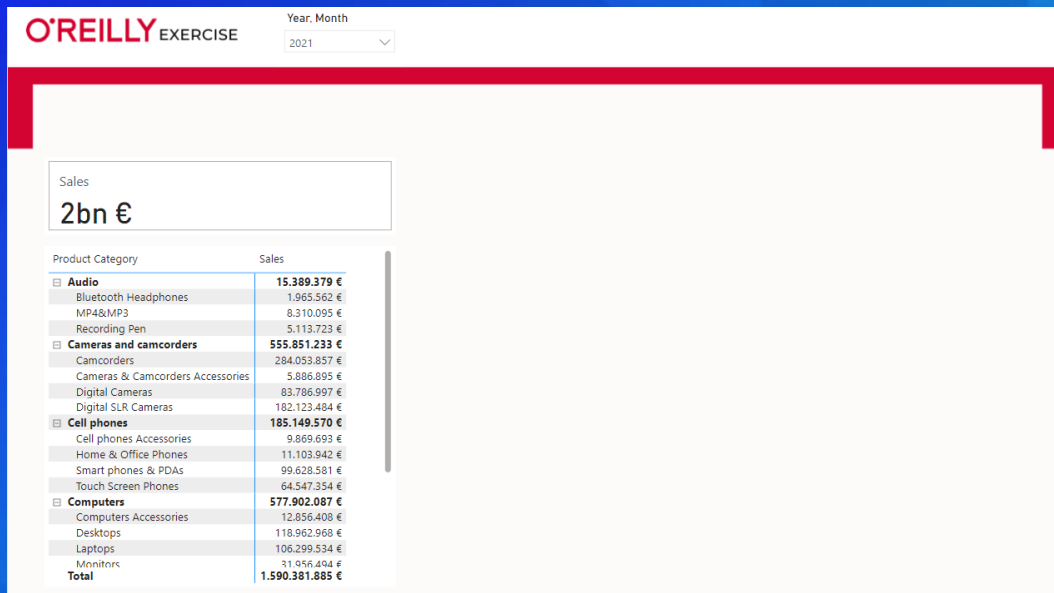
Cancel

EXERCISE 1: Table, matrix, card & slicer



Open file “1. GettingStartedWithPowerBI.pbix”

Follow the instructor's step-by-step guidance to create these visuals:





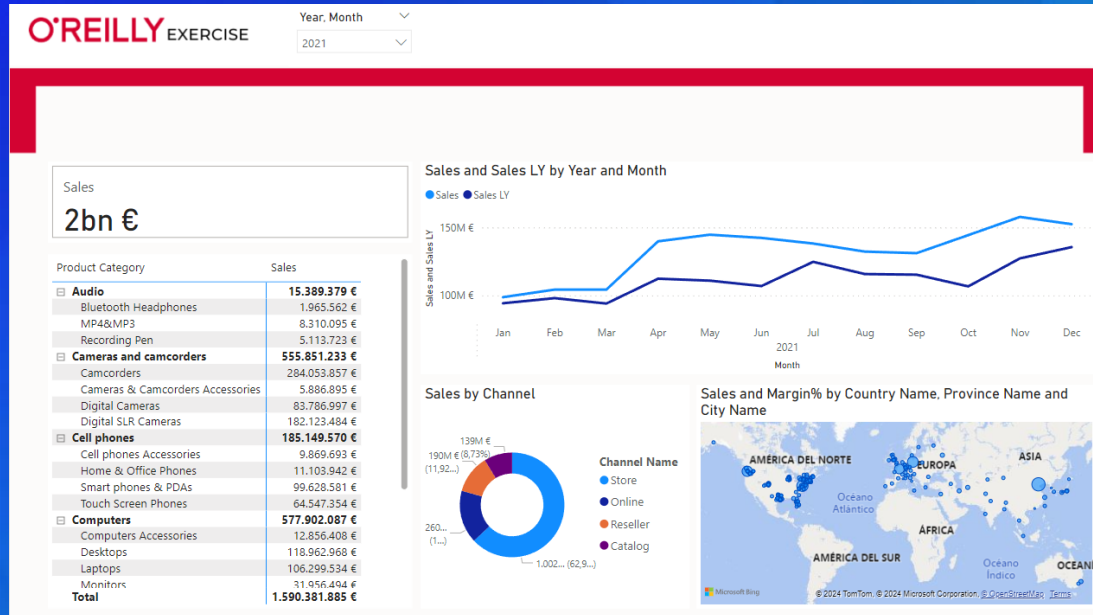
DEMO 2: Bar, Pie, Line and Maps.

EXERCISE 2: Bar, Pie, Line and Maps



Open file "2. GettingStartedWithPowerBI.pbix"

Follow the instructor's step-by-step guidance to create these visuals:





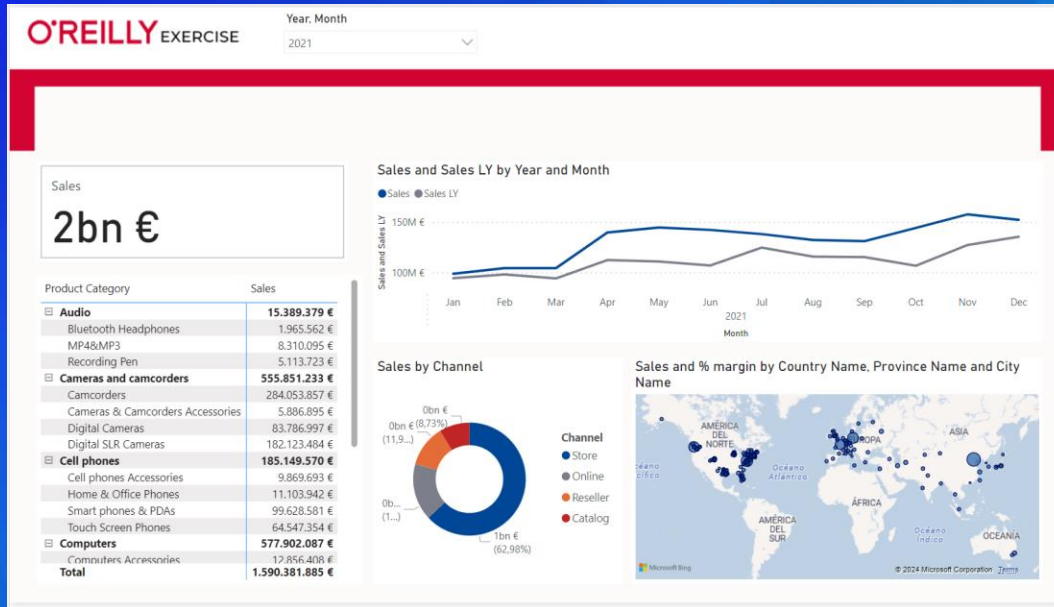
DEMO 3: Background and theme of report

EXERCISE 3: Background and theme of report



Open file “3. GettingStartedWithPowerBI.pbix”

Follow the instructor's step-by-step guidance to change the theme:





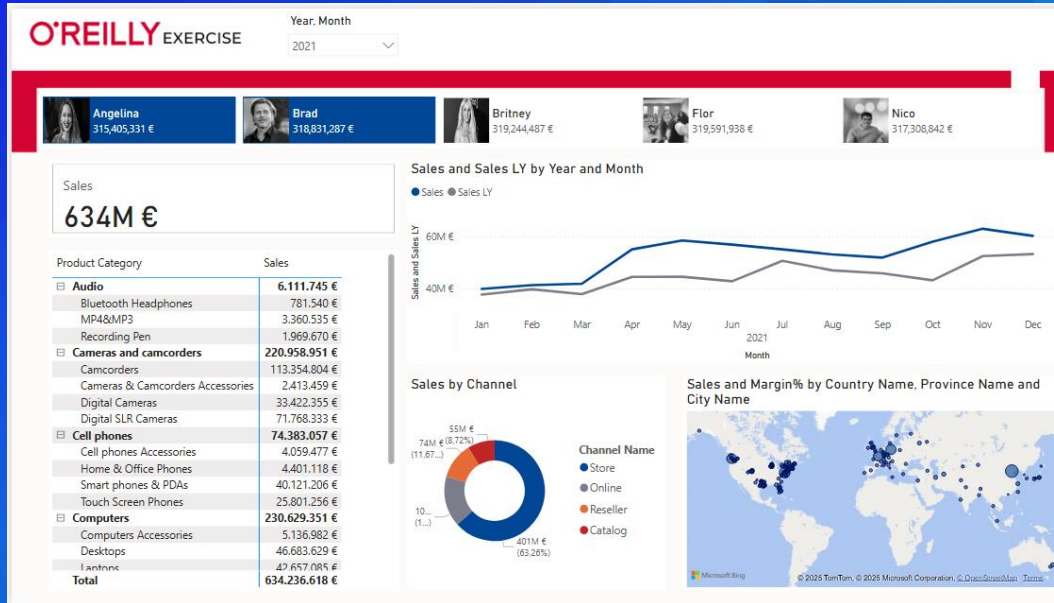
DEMO 4: Slicer

EXERCISE 4: Slicer



Open file “4. GettingStartedWithPowerBI.pbix”

Follow the instructor's step-by-step guidance to create these visuals:





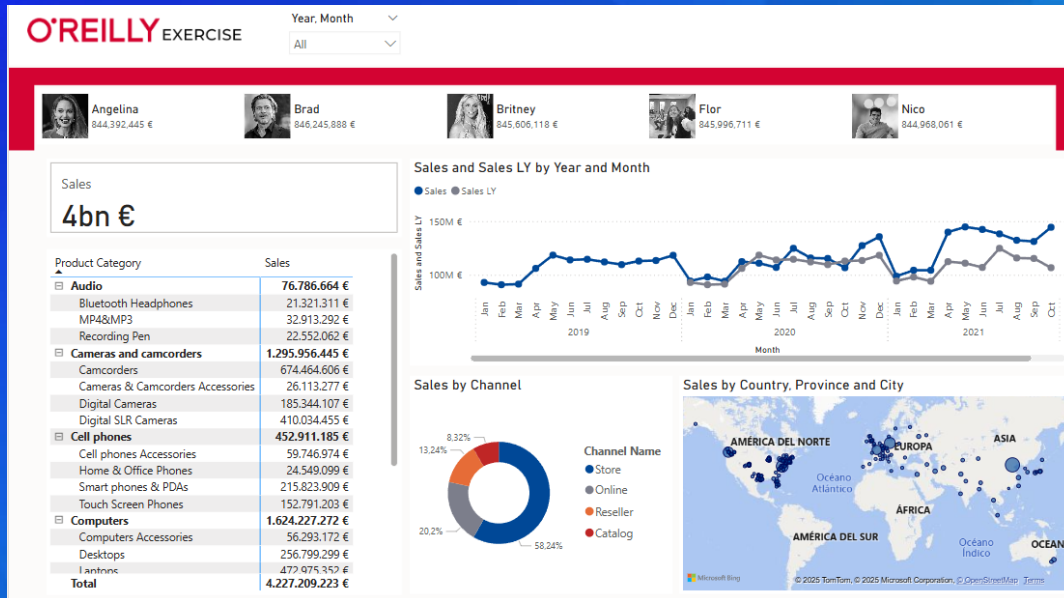
DEMO 5: Format Pane

EXERCISE 5: Format Pane



Open file “5. GettingStartedWithPowerBI.pbix”

Follow the instructor's step-by-step guidance to create these visuals:





Power BI Service

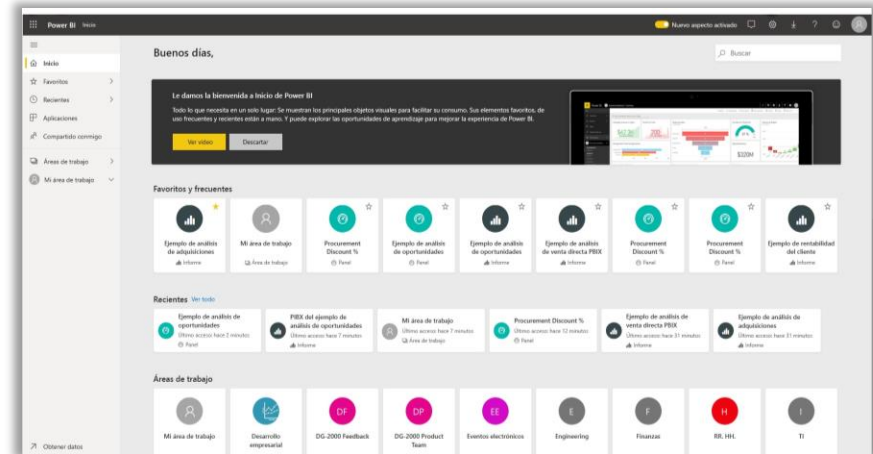




Introduction to Power BI Service

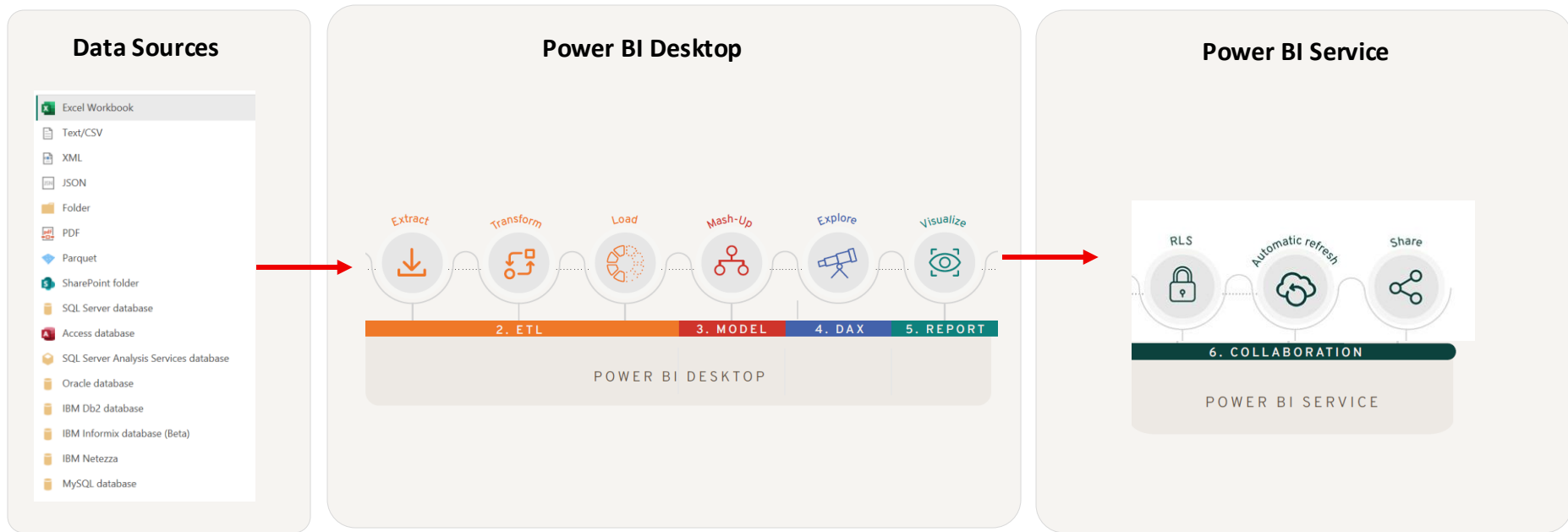
Power BI is a cloud-based business analytics service that allows anyone to:

- Visualize and analyze data faster, more efficiently and more consciously
- Connect users to a wide variety of data
- Use easy-to-use dashboards, interactive reports and engaging visualizations that bring data to life
- Insights from any device
- Collaboration across your organization

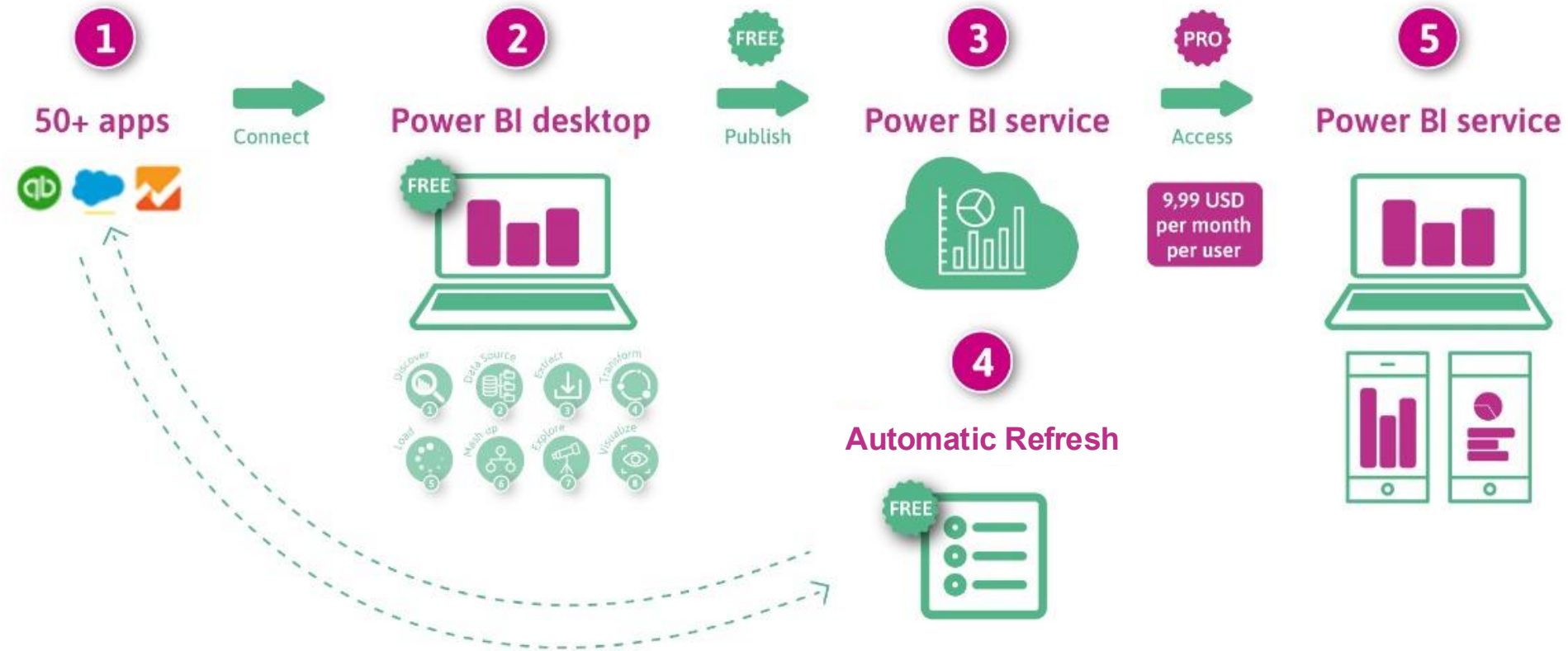




Introduction to Power BI Service



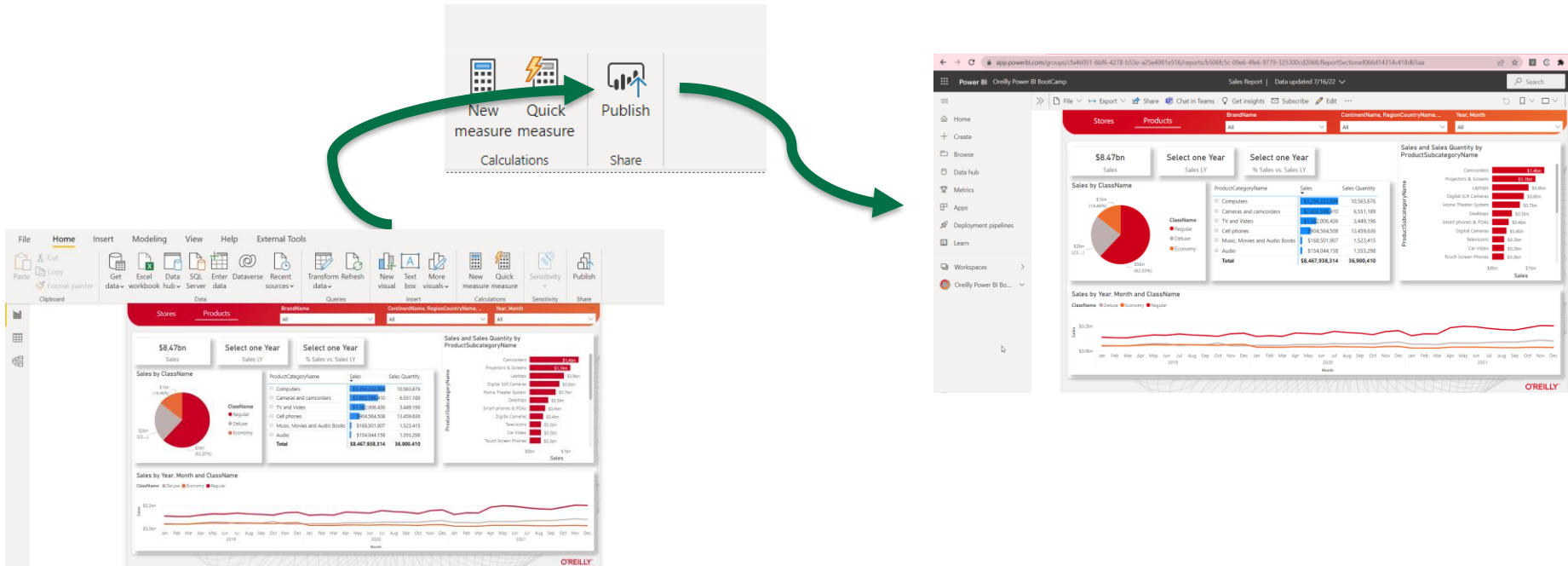
Power BI architecture



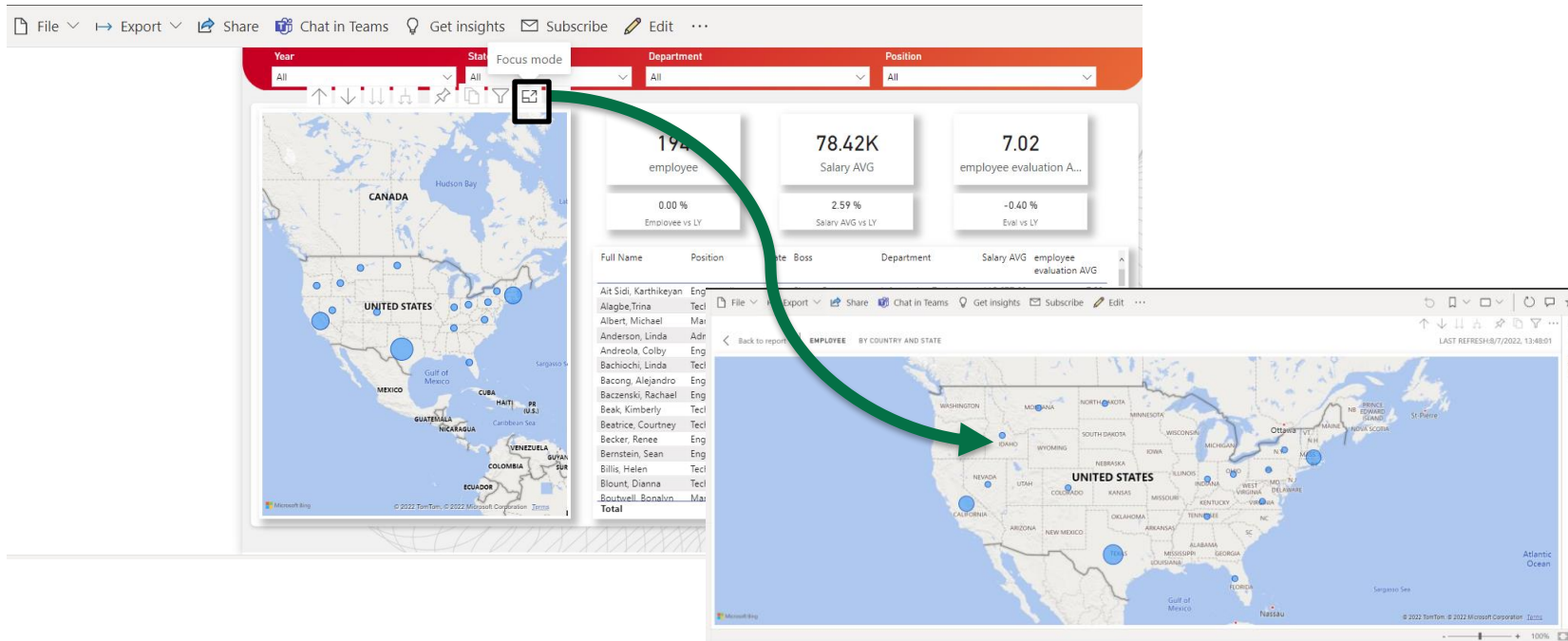


Introduction to Power BI Service

Before we start, we are going to publish the report created in the previous module. We do this through Power BI Desktop using the Publish button.



Report view



Report view



File Export Share Chat in Teams Get insights Subscribe Edit

Year: All State: All Department: All Position: All

194 employee 78.42K Salary AVG 7.02 employee evaluation A...

0.00 % Employee vs LY 2.59 % Salary AVG vs LY -0.40 % Eval vs LY

Full Name Position State Boss Department

Alt Sidi, Karthikeyan	Engineer II	MA	Simon Roup	Information Techn
Alagbe, Trina	Technician I	CA	Elijah Gray	Production
Albert, Michael	Manager	MA	Janet King	Production
Anderson, Linda	Administrative II	TX	David Stanley	Admin Offices
Andreola, Colby	Engineer II	CA	Alex Sweetwater	Software Engineer
Bachiochi, Linda	Technician I	TX	Brannon Miller	Production
Bacong, Alejandro	Engineer I	TX	Peter Monroe	Information Techn
Baczinski, Rachael	Engineer I	TX	Alex Sweetwater	Software Engineer
Beak, Kimberly	Technician II	CA	Kelley Spirea	Production
Beatrice, Courtney	Technician II	CA	Elijah Gray	Production
Becker, Renee	Engineer I	MA	Simon Roup	Information Techn
Bernstein, Sean	Engineer I	TX	Alex Sweetwater	Software Engineer
Billis, Helen	Technician I	TX	Brannon Miller	Production
Blount, Dianna	Technician II	MA	Michael Albert	Production
Boutwell, Bonalyn	Manager	CA	Elijah Gray	Admin Offices
Total				

Full screen Fit to page Fit to width Actual size High-contrast color

194 employee 78.42K Salary AVG 7.02 employee evaluation A...

0.00 % Employee vs LY 2.59 % Salary AVG vs LY -0.40 % Eval vs LY

Full Name Position State Boss Department Salary AVG employee evaluation AVG

Alt Sidi, Karthikeyan	Engineer II	MA	Simon Roup	Information Technology	115,677.60	7.33
Alagbe, Trina	Technician I	CA	Elijah Gray	Production	37,800.00	8.67
Albert, Michael	Manager	MA	Janet King	Production	100,389.00	6.00
Anderson, Linda	Administrative II	TX	David Stanley	Admin Offices	61,789.00	8.67
Andreola, Colby	Engineer II	CA	Alex Sweetwater	Software Engineering	67,904.33	9.67
Bachiochi, Linda	Technician I	TX	Brannon Miller	Production	45,841.33	7.00
Bacong, Alejandro	Engineer I	TX	Peter Monroe	Information Technology	86,213.42	8.00
Baczinski, Rachael	Engineer I	TX	Alex Sweetwater	Software Engineering	88,416.00	5.67
Beak, Kimberly	Technician II	CA	Kelley Spirea	Production	86,942.40	7.67
Beatrice, Courtney	Technician II	CA	Elijah Gray	Production	74,983.75	8.00
Becker, Renee	Engineer I	MA	Simon Roup	Information Technology	81,558.95	5.33
Bernstein, Sean	Engineer I	TX	Alex Sweetwater	Software Engineering	88,416.00	5.33
Billis, Helen	Technician I	TX	Brannon Miller	Production	30,789.33	8.00
Blount, Dianna	Technician II	MA	Michael Albert	Production	71,246.50	7.33
Boutwell, Bonalyn	Manager	CA	Elijah Gray	Admin Offices	66,307.58	7.00
Total					78,422.33	7.02

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Export data to PDF, PowerPoint



File Export Share Chat in Teams Get insights Subscribe Edit ...

Analyze in Excel
PowerPoint
PDF

Year: All State: All Department: All Position: All

194 employee
0.00 % Employee vs LY

78.42K Salary AVG
2.59 % Salary AVG vs LY

7.02 employee evaluation AVG
-0.40 % Eval vs LY

Map of the United States showing data points across various states.

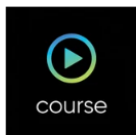
Full Name	Position	State	Boss	Department	Salary AVG	employee evaluation AVG
Ait Sidi, Karthikeyan	Engineer II	MA	Simon Roup	Information Technology	115,677.60	7.33
Alagbe, Trina	Technician I	CA	Elijah Gray	Production	37,800.00	8.67
Albert, Michael	Manager	MA	Janet King	Production	100,389.00	6.00
Anderson, Linda	Administrative II	TX	David Stanley	Admin Offices	61,789.00	8.67
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Bacong, Alejandro	Engineer I	TX	Peter Monroe	Information Technology	86,213.42	8.00
Baczenski, Rachael	Engineer I	TX	Alex Sweetwater	Software Engineering	88,416.00	5.67
Beak, Kimberly	Technician II	CA	Kelley Spira	Production	86,942.40	7.67
Beatrice, Courtney	Technician II	CA	Elijah Gray	Production	74,083.75	8.00
Becker, Renee	Engineer I	MA	Simon Roup	Information Technology	81,558.95	5.33
Bernstein, Sean	Engineer I	TX	Alex Sweetwater	Software Engineering	88,416.00	5.33
Billis, Helen	Technician I	TX	Brannon Miller	Production	30,789.33	8.00
Blount, Dianna	Technician II	MA	Michael Albert	Production	71,246.50	7.33
Boutwell, Bonalyn	Manager	CA	Elijah Gray	Admin Offices	66,307.58	7.00
Total					78,422.33	7.02

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More Resources

- Video course: [“Power BI Fundamentals: Building a Strong Foundation in Data Analysis”](https://learning.oreilly.com/course/power-bi-fundamentals/0636920986317/)



Power BI Fundamentals: Building a Strong Foundation in Data Analysis

By [Maria Florencia Hourcouripé](#) and [Nicolás Lagreste Zucchini](#)

[O'Reilly Media, Inc.](#) • February 2024

Write the [first review](#) ⌚ 5h 14m 📝 Includes quizzes

<https://learning.oreilly.com/course/power-bi-fundamentals/0636920986317/>



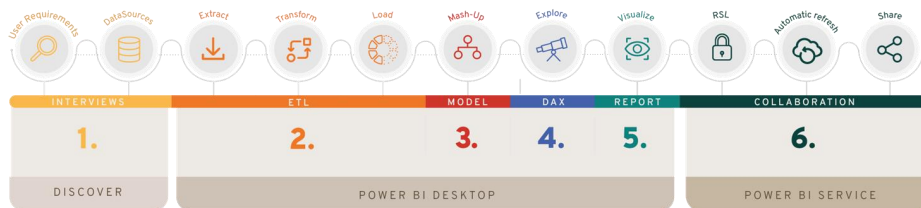


Recommendations for Next Steps

Power BI Bootcamp Series

Advanced Learning Path:

- 1) Power BI Bootcamp: Building the Foundations for Power BI
- 2) Power Query: Data Connections and Transformations
- 3) Data Modeling for optimal structure & DAX for calculations
- 4) Visualizations for impactful reporting & Power BI Service





Stay Connected

Follow us on LinkedIn:



Nicolás Lagreste Zucchini



María Florencia Hourcouripé

Let's Collaborate: Interested in a personalized mentorship or consulting session? Contact us:

-  nico@analyticmood.com
-  flor@analyticmood.com

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- www.analyticmood.com for resources, upcoming events, and blog articles.

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Thank you!